

OTA Wagon Salient Characteristics

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Wagon:

- Outward aesthetic shall match current Caisson wagon design
- Caisson must have six (6) stars on either side of the caisson
- The length of the caisson, limber box, and limber pole shall be 320" long
- Limber pole shall be 165"
- Limber box shall be 50" long x 49" wide
- Limber box seat shall be 24" in height
- Caisson shall be 115" length x 44" wide x 18" height
- Wheels shall be 55" in height

Limber Design and Functionality:

- Adjustability: The limber features adjustable height settings with the ability to adjust and lock into positions between 33" to 47" from the ground (measured from the end of the limber pole) to account for different horse heights. Adjustment increments shall be no larger than 1.5" (measured at the limber pole fulcrum).
- Detachment: The limber easily detaches from the caisson wagon at the support hitch. The limber pole can be detached separately at the support hinges. Limber shall have the ability to swivel left, right, up, and down separately from the Caisson with turns and hills.
- Seat: The Limber Seat shall be sturdy while attached, and detachable for repairs.
- Spreader Bar: Spreader bar shall be easily connected to the limber pole with the use of carabiners. Each wagon shall come with two spreader bars painted black.
- Limber Pole: Limber pole shall be made of lightweight durable metal, round, and painted black. Each wagon shall come with two limber poles. One limber pole shall feature a curved dip to provide space for wheel rider's leg while riding, the other limber pole shall be straight. Dipped limber pole shall be adjustable at least 6 inches to accommodate different sized wheel horses. Limber pole shall have the ability to swivel left and right separately from the limber box with the movement of the horses' steps as they pull the Caisson.
- Limber Box: Limber box shall be weather resistant and sound dampening. Limber box shall be a smooth box design. Limber box seat shall be able to open and store

equipment. Storage must be sound dampened and lined with carpet like material. The storage space must be a minimum of 20 cubic feet and a minimum depth of 12 inches.

Caisson Wagon Design and Functionality:

- **Wheel Design:** Minimum of 3" wide wheels are used. Wheels shall have durable, solid rubber smooth tread with a minimum vertical thickness of 1" composed of hard rubber or similar material to dampen the sound of the wagon noise while being driven on pavement. Wheels shall be 55" tall. Hubs shall be flush with the wheel and have the outward appearance of brass. Provide one spare wheel for each wagon. If the limber and caisson wheels differ, provide one spare of each type per wagon.
- **Parking Brake System:** Drum brakes or similar are employed for forward movement. Parking brake system shall not be obviously outwardly visible. Shall be able to be applied by wheel rider while riding.
- **Surge Brake System:** A specialized brake system (such as surge brake or similar) automatically engages brakes to assist with downhill braking to reduce load on wheel horses. Brake system shall act similar to a common boat trailer. Shall be able to be disengaged in case of a malfunction and shall not be outwardly visible.
- **Casket Rollers:** Wagon shall have casket rollers to ease loading and unloading of the casket, protect the top of the wagon and bottom of the casket.
- **Caisson Box:** The Caisson Box shall have a protective cover on top edge of the back of the Caisson to protect the box from being damaged in the process of loading a casket. Additionally, Caisson box shall have a holder for the stop clamp for storage when not in use.
- **Caisson Kickstand:** Kickstand shall be installed so that when the Caisson is detached from the limber box, the Caisson will be supported off the ground by the kickstand. The kickstand shall be able to be locked into position (up when not in use and down when in use).
- **Sign Hook:** A hook to hang a sign off the back of the Caisson shall be emplaced underneath and towards the rear of the Caisson. The hook shall not be outwardly visible, but a sign hanging from the hook shall be visible to personnel behind the Caisson.

Additional Wagon Considerations:

- **Materials:** Frame and other structural components shall be constructed of powder-coated high-grade metal material for durability. Outward aesthetic must match those as outlined in the previous models document and shall be constructed with high-grade metal material, besides wheels.
- **Weight Limit:** Overall weight shall not exceed 2000 lbs.
- **Mileage:** Wagon shall be able to sustain daily use and mileage of a minimum 10 miles per day @ 10 mph on 10% grade hills. Must be able to carry an 800lb casket.
- **Maintenance Program/Lifecycle:** Maintenance program is a written program the Government can follow and implement and shall cover a minimum of any moving or load bearing parts on the wagon (i.e. rollers, bearings, bolts, hubs, chassis, brakes, suspension, etc.). Provide a maintenance program based on age and mileage. Expectation is that the wagon lasts a minimum of 40 years with assistance of the maintenance program. Unit has access to wheeled vehicle mechanics, welders, leather workers, etc. Repair parts shall be easily resourced commercially.
- **Hub Maintenance:** Hubs shall be designed such that maintenance is roughly 1x per 6 months (other than necessary greasing).
- **Load Distribution On Horses:**
 - When stationary on flat ground: There shall be no load from wagon components on horses.
 - When moving: Load shall be applied to the traces, with line of draft engaging horses' breast collar.
 - When stopping: Weight of the load shall be minimized by the braking system, with residual weight applied to the horses' breeching. The height of the limber pole shall be minimally affected by the stopping of the Caisson. The stopping of the caisson shall cause the limber pole to move in a downward motion.
 - When going downhill: With assistance from the surge brake, the weight distributed on the horses shall be the same as when stopping. The height of the limber pole shall be minimally affected by the stopping of the Caisson. The downhill weight of the caisson shall cause the limber pole to move in a downward motion.
- **Suspension System:** The limber utilizes a suspension system to withstand the Daily Mileage (listed below) and is balanced while riding, supporting weight distribution

needs. Suspension system shall provide a minimum of 2" of travel. Suspension shall be easily adjustable without crawling under the wagon.

- Odometer: Wagon shall include an odometer. Odometer shall not be outwardly visible.

Warranty: Contractor shall include a five-year warranty to cover manufacturer deficiencies of the entire wagon.